

ObsSec Recommendations for Michaelmas 2026

Introduction

This document contains a selection of our recommended objects to look at with the telescopes this summer. It is by no means a comprehensive list of everything there is to see, but instead contains some of the brighter or otherwise interesting objects, and provides a good starting point for visual observing. We've included no photos of anything, since these often make your own observations pale in comparison and distract you from appreciating that you can see the objects with your *very own eyes*.

The selection of objects is organised by constellation, which are themselves organised roughly in increasing Right Ascension (RA). As a result, objects listed earlier in the guide are highest in the sky and best seen earlier in the term or earlier in the evening. Objects near the end are best seen later in the term or later at night.

For each object, there is:

- The catalog name [*Messier (M)*, *Index (IC)*, *New General (NGC)*] and any common name.
- Object class, described below.
- Our difficulty rating, explained below.
- The apparent magnitude, which corresponds to how bright the object is. A smaller magnitude means brighter, and any object brighter than ~5 can be seen with the naked eye with our levels of light pollution.
- Celestial coordinates, in RA and Dec (Declination).
- A brief description of the object, with some observing tips and specific points of interest.

Difficulty scale:

- **Easy** ▾
These objects are great to start with. They will typically be easy to find, have lots of visible features, and will jump out of the eyepiece at you.
- **Medium** ▾
These are harder to spot, requiring some effort to find and especially to make out detail. Moonless nights and turning off the dome lights will help.
- **Hard** ▾
Most of these objects need some experience to observe, including long dark adaptation and averted vision. Detail can be very tricky to discern, and often a successful observation reveals just a faint blur and the knowledge that you've seen something really exciting with your own eyes.
- **Insane or Astrophotography** ▾
These objects might well be impossible to see in Cambridge, requiring larger telescope apertures and darker skies than we have here. Do give them a go, however, and many of them are also great targets for astrophotography if you have the ability to do that.

Object Classes:

- **Planets**
Large bodies orbiting our sun. You are standing on one of them! Through a telescope, they are small and bright, so take magnification (via smaller eyepieces) well. Details on their surfaces are limited by seeing (atmospheric turbulence) instead of contrast (from light pollution).
- **Multiple star systems**
Stars that are physically close to and orbit each other. The contrast between different star types is readily apparent for closely separated ones. Some particularly tight pairs can require good seeing to 'split' (i.e. to convince yourself that the point of light is actually two stars).
- **HII regions**
Large glowing emission nebulae, primarily of hydrogen gas. These act as stellar nurseries, and the newly formed stars within them ionise the gas to glow a deep red, among other colours (which are usually only visible in photos). Often only unusually bright regions are visible visually, although filters can help increase the contrast.
- **Planetary nebulae**
Named for being small, round, relatively bright, and therefore planet-like, these emission nebulae are glowing shells of gas surrounding hot dying stars. Their small size and higher contrast make them great targets for teasing out difficult detail in a large focal length telescope like the Northumberland.
- **Supernova remnants**
Emission nebulae formed from the violent explosion of the progenitor star at the end of its life. While rare and (astronomically) short-lived, we do have the *Crab Nebula* (M1) visible with the telescopes, appearing similar to a faint planetary nebula.
- **Reflection nebulae**
Often combined with other nebula types, these are simply dust clouds reflecting broadband starlight. As a result, they appear white in photographs and are not improved with filters, sometimes necessitating darker skies than we have here.
- **Open clusters**
Groups of stars which all formed together, and younger ones are often associated with the HII region that birthed them. They appear as gatherings of tens to hundreds of easily distinguishable points.
- **Globular clusters**
Collections of thousands to hundreds of thousands of gravitationally bound stars. These are often old and distant, outside the plane of our galaxy. They can be seen as hazy globes, in which you can often discern the faint speckling of the thousands of constituent stars.
- **Galaxies**
These are the most distant objects easily seen, and are collections of tens of billions of stars and all the other objects discussed here. They are usually visible through the telescopes as elliptical clouds with a bright core, and some have subtle detail from spiral arms or dust lanes that can be picked out with practice.

Good luck observing, and clear skies!

- *The ObsSecs, 2026*

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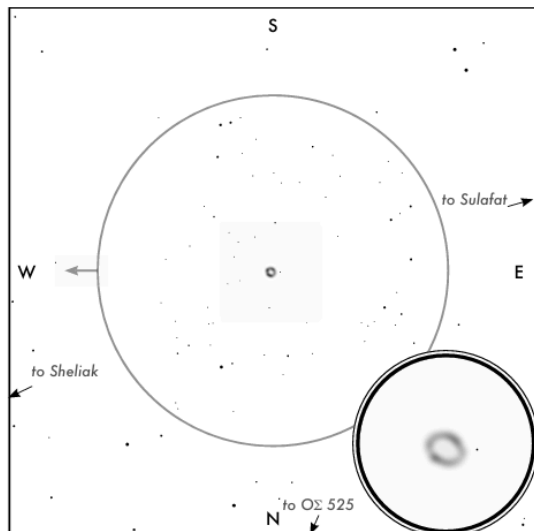
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Photo credit: Stellarium. Shown is the Michaelmas sky from Cambridge, looking southwards. Note the bright disk of Saturn on the border of the constellations Pisces and Cetus, and the Andromeda Galaxy as the circled object in Andromeda.

Note: Unless otherwise specified, figures have been taken from ‘*Turn Left at Orion*’ - G. Consolmagno, D. M. Davis. Object descriptions were written by William Moorhouse (ObsSec 2026-27).

Lyra



M57 - Ring Nebula (Planetary Nebula) Easy ▾

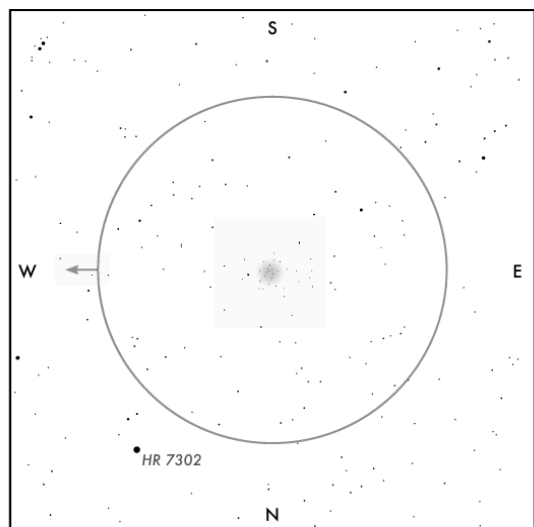
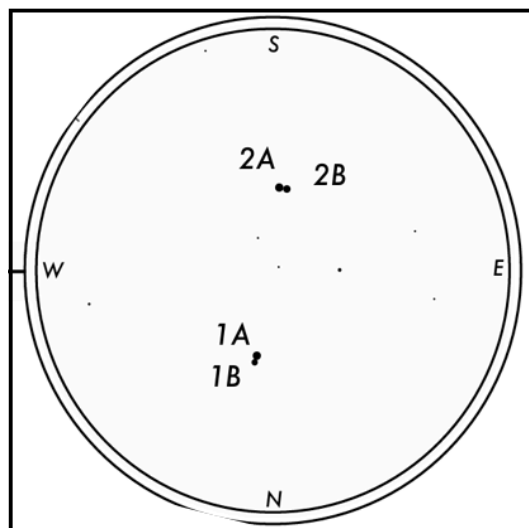
RA 18h 54m 35s | Dec 33° 03' 38" | Mag 8.80

As a planetary nebula, M57 (distant ~ 2,500 light years from Earth) is the result of the death of a low mass star forming a white dwarf. The UV radiation from the white dwarf ionises the gas, which then emits the light we observe. The distinct ring structure comes from the much higher density of oxygen and nitrogen in the outer layers compared with the helium core surrounding the white dwarf. (It is the concentrated levels of O III which dominates the surface brightness of the outer layer.) Expanding at 12 miles per second, it would have taken ~ 20,000 years for the nebula to reach its current size.

Epsilon Lyrae - Double Double (Multiple Star) Easy ▾

RA 18h 45m 13s | Dec 39° 41' 49" (Epsilon1 Lyrae)

With Vega (Alpha Lyrae) in the finder scope, observe a triangle of bright stars with Zeta Lyrae south of Vega and Epsilon Lyrae in the northwest. Separated by 209", ϵ Lyr can be easily resolved in the finder, but its two orbiting components, ϵ^1 Lyr and ϵ^2 Lyr, require higher magnifications. The components of both ϵ^1 and ϵ^2 Lyr are separated ~ 150 AU (2.2" and 2.4", respectively). The heavier components of ϵ^2 Lyr take ~ 600 years to orbit their centre of mass, whilst it takes more than 1000 years for ϵ^1 Lyr. The system is 160 light years distant, and consists of white stars of spectral type A.



M56 (Globular Cluster) Medium ▾

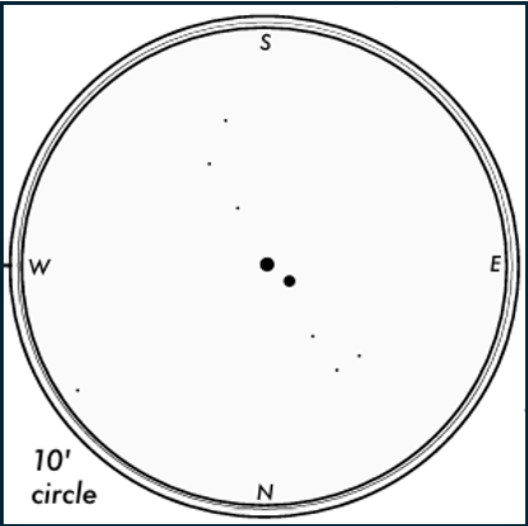
RA 19h 17m 38s | Dec 30° 13' 56" | Mag 8.39

Located roughly halfway between Albireo and Sulafat, M56 is one of the fainter Messier globulars and can be a difficult find. It lacks the characteristic bright core of most other globulars, but still contains more than 100,000 stars and measures ~ 85 light years across (8.8'). Sitting 32,900 light years away in a retrograde orbit through the Milky Way, it is thought that M56 was formed from a dwarf galaxy merger. The largest globular cluster, Omega Centauri (visible only in the southern hemisphere), would then form the surviving nucleus of this capture of a dwarf galaxy by the Milky Way.

In the constellation Lyra, you might also like to observe:

- Sheliak (Double Star) Easy ▾
- T Lyrae (Carbon Star) Medium ▾
- Struve 2470 and Struve 2474 ((Optical) Double Double Star System)) Medium ▾

Cygnus



Beta Cygni - Albireo (Double Star) Easy ▾

RA 19h 31m 48s | Dec 28° 00' 55" | Mag 3.19, 4.67

Star	Magnitude	Colour	Location
A	3.19	Yellow-orange	Primary star
B	4.67	Blue-white	34" NE from A

Albireo is made of a giant, orange (spectral type K) star orbited by a hot, blue (spectral type B) star. Separated by at least 4000 AU, the companion takes ~ 100,000 years to orbit the primary - far too slow to observe any motion.

M29 - Cooling Tower Cluster (Open Cluster) Easy ▾

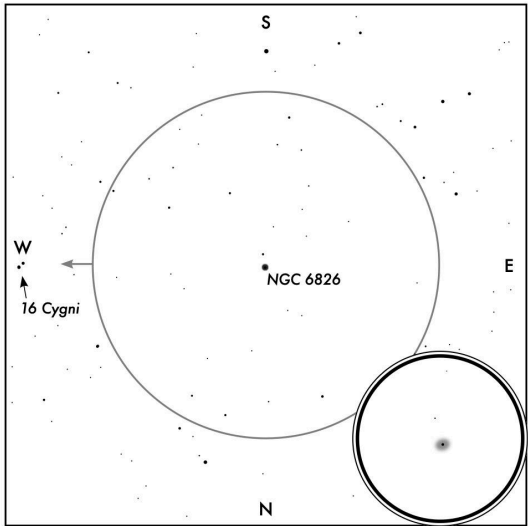
RA 20h 24m 56s | Dec 38° 35' 10" | Mag 6.59

Just south of Sadr (the central bright star in Cygnus) and within the nebulous, spiral arms of the Milky Way, is M29. Its five brightest members resemble the Plough asterism (a 'stubby dipper'), and are hot, blue stars of spectral class B0. The cluster is relatively young, ~ 10 Myr, and distant ~ 5000 light years.

M39 (Open Cluster) Easy ▾

RA 21h 32m 46s | Dec 48° 33' 02" | Mag 4.59

Located ~ 9° NE of Deneb, M39 is a loosely-structured open cluster that is best observed at low powers (it spans a similar size to a full Moon). All of its members are main sequence stars. With the brightest at the point before evolution into the red giant phase, M39 has an estimated age of 270 Myr.



NGC 6826 - Blinking Planetary Nebula Hard ▾

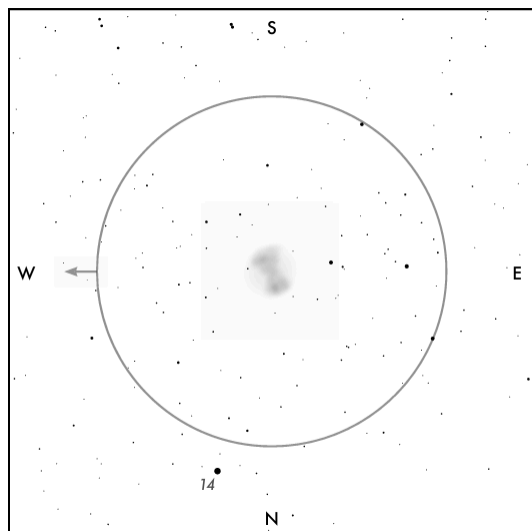
RA 19h 45m 31s | Dec 50° 35' 20" | Mag 8.89

The nebula shows clearly with averted vision and may disappear entirely if you look at it directly, since its light (equivalent to a ninth-magnitude star) is spread out over a small disk. Looking at it directly, you'll only be able to see the central star - a white dwarf left over from the collapse of a red giant. This "here again, gone again" feature of the nebula as the observer's eye wanders, gives it its name. It is located about 2000 light years away, and measures roughly 15,000 AU in diameter. It is a difficult find with the Dobsonian or Thorowgood, but can be reached with some star hopping.

In the constellation Cygnus, you might also like to observe:

- 16 Cygni (Double Star) Easy ▾
- NGC 6960 - Veil Nebula (Supernova Remnant) Insane or Astrophotography ▾ (O III filter)
- NGC 6888 - Crescent Nebula (Emission Nebula) Insane or Astrophotography ▾ (O III filter)

Vulpecula, Sagitta, Delphinus



M27 - The Dumbbell (Planetary Nebula) Medium ▾

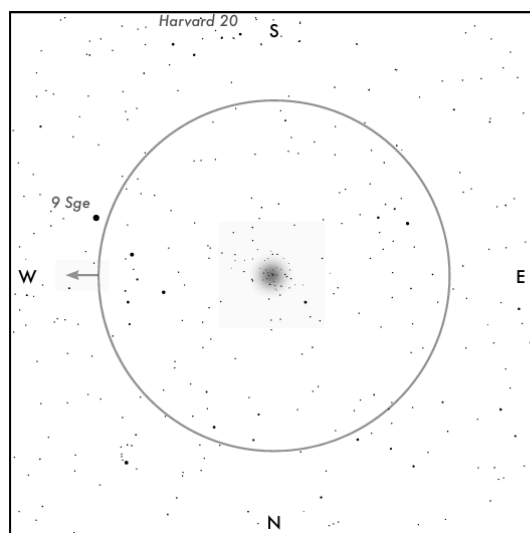
RA 20h 00m 45s | Dec 22° 47' 22" | Mag 7.09

M27 is formed from the collapse of a formerly Sun-like star to a white dwarf. The expanding gas shell, formed from the outer layers of the central star, ballooned into a pair of lobes above and below the plane of a ring of dense gas the star had previously ejected. M27 has a prolate spheroidal shape which, when viewed along the plane of the equator, means the two lobes are seen. Looking along the poles, M27 might bear a ring shape similar to M57 in Lyra. Extending two light years in diameter and expanding at a rate of 30 kilometers per second, the nebula has a kinematic age of 10,000 years.

M71 (Globular Cluster) Medium ▾

RA 19h 54m 58s | Dec 18° 50' 54" | Mag 8.39

For many decades, M71 was considered a densely packed open cluster, similar to M11 in Scutum. Its lack of RR Lyrae variable stars (used as standard candles) and high metallicity made the case for this, until the advent of modern photometry. The relative youth of M71, aged between 9 - 10 Gyr, also explains the abundance of 'metals'. (Most globulars formed from a giant molecular cloud in the very early universe before the first stars were able to go supernova, producing heavier elements.) Distant 13,000 light years, M71 can be easily located halfway between Gamma and Delta Sagittae.



Collinder 399 - Brocchi's Cluster (Asterism) Easy ▾

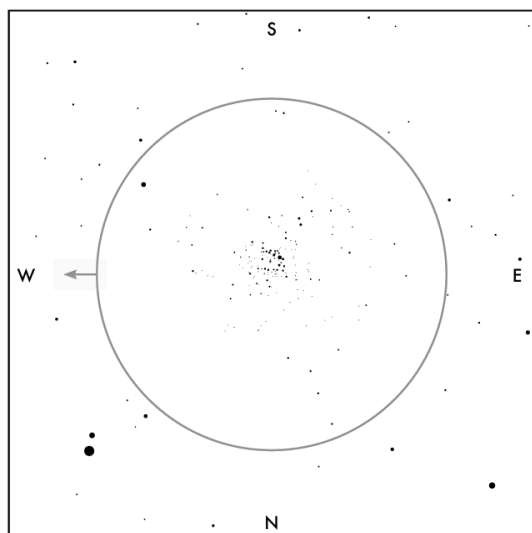
RA 19h 26m 34s | Dec 20° 14' 09" | Mag 3.59

At an apparent size of 89.0' (roughly three full Moons), Brocchi's Cluster is best viewed using binoculars or a finder scope and is a collection of fifth- to seventh-magnitude stars. Once thought to be an open cluster of gravitationally bound stars, it was not until 1997, when the Hipparcos satellite made measurements of stellar distances and proper motions, that the cluster was found to be a chance grouping of unrelated stars. Ranging from 200 to 2000 light years distant, the stars can be located a third of the way along from Altair to Vega. (Photo credit: Cloudy Nights 22/10/2007 WadeVC.)

In the constellations Vulpecula, Sagitta and Delphinus, you might also like to observe:

- Gamma Delphini ((Visual) Double Star) Easy ▾
- NGC 6823 (Open Cluster) Easy ▾ within NGC 6820 (Emission Nebula) Astrophotography ▾
- NGC 6934 (Globular Cluster) Hard ▾

Scutum, Aquila, Draco



M11 - The Wild Ducks (Open Cluster) Medium ▾

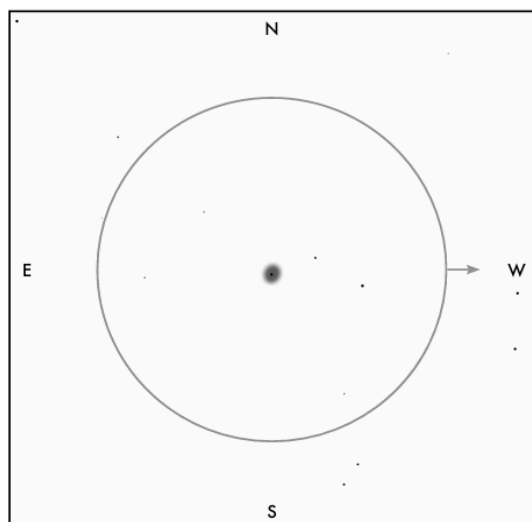
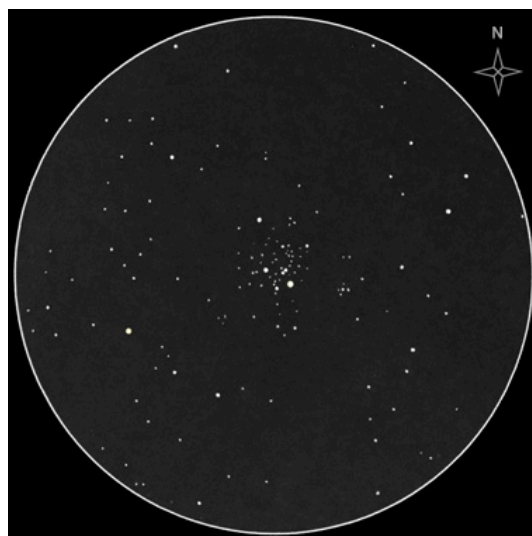
RA 18h 52m 31s | Dec -06° 14' 07" | Mag 5.80

M11 is best viewed at medium power (12 - 18 mm eyepiece) in a Dobsonian, as this will resolve the cluster into hundreds of individual stars against a background of grainy light. Only ~ 100 light years across with a core radius of 4.0 light years, the stars within M11 are, on average, less than one light year apart, making it one of the most compact open clusters. The brightest members are hot, young, blue and white stars of spectral class A and F. However, over a dozen stars have evolved to the red-giant stage, suggesting the cluster (6000 light years distant) is a few hundred millions years old.

M26 (Open Cluster) Medium ▾

RA 18h 46m 46s | Dec -09° 21' 21" | Mag 8.00

One field of view in the finder scope southwest of M11, is the smaller and dimmer open cluster, M26. Distant 5000 light years and spanning 22 light years across, M26 is a collection of ~ 30 tenth-magnitude, and fainter, stars. Notably, there is a region of low star density near the nucleus of the cluster. This was thought to be the result of an obscuring cloud of interstellar matter between Earth and M26, but a paper, late last century, claimed that this was not possible. M26 is aged at 85.3 million years, and is best observed at medium power in a Dobsonian. (Photo credit: [deepskywatch.com](https://www.deepskywatch.com).)



NGC 6543 - Cat's Eye Nebula (Planetary Nebula) Hard ▾

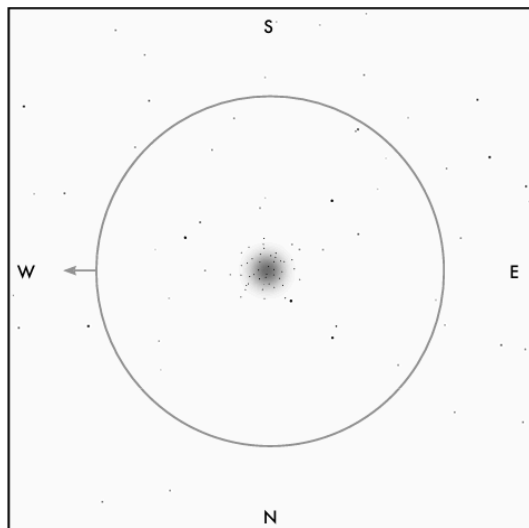
RA 17h 58m 33s | Dec 66° 37' 48" | Mag 8.10

NGC 6543 is a relatively young, yet greatly studied, planetary nebula, distant 3300 light years, with a highly complex structure of knots, jets and concentric gas shells. These shells that have been ejected by the central star over the past thousand years give resemblance to a cat's eye, though are difficult to discern visually. With a high Northern declination, refractor telescopes, such as the Northumberland and Thorowgood, are not suitable. Observed through a Dobsonian, say, NGC 6543 has a distinct blue-green colour and is best viewed using averted vision.

In the constellations Scutum and Aquila, you might also like to observe:

- V Aquilae (Variable, Carbon Star) Medium ▾
- NGC 6712 (Globular Cluster) Medium ▾
- Delta Scuti (In 1.25 million years, it will be 9 light years from Earth, shining as bright as Sirius.)

Aquarius



M2 (Globular Cluster) Medium ▾

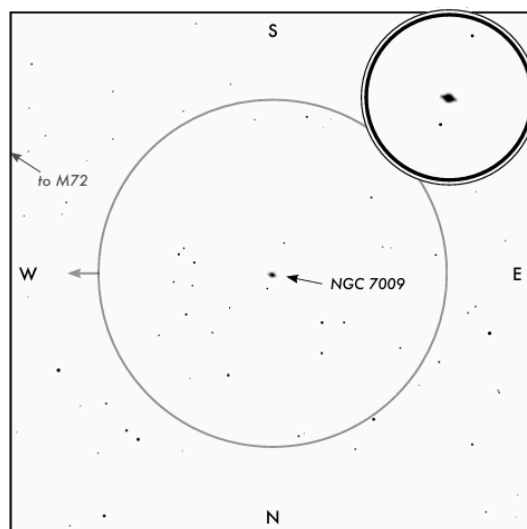
RA 21h 34m 50s | Dec -00° 42' 15" | Mag 6.59

With a diameter of 175 light years, M2 is one of the largest globular clusters known. At medium power in a Dobsonian, it is distinctly elliptical in the northwest-southeast direction with a modestly intense core and many individual stars that can be resolved across the disk. Using averted vision, several dark lanes (due to obscuring clouds of interstellar dust) may be observed, most prominently, in the NE of the cluster. Distant 37,500 light years, M2 also has an estimated age of 13.2 billion years (Botev et al., 2025) and is part of the Gaia Sausage (the remains of a dwarf galaxy merger).

NGC 7009 - Saturn Nebula (Planetary Nebula) Hard ▾

RA 21h 05m 38s | Dec -11° 15' 34" | Mag 7.80

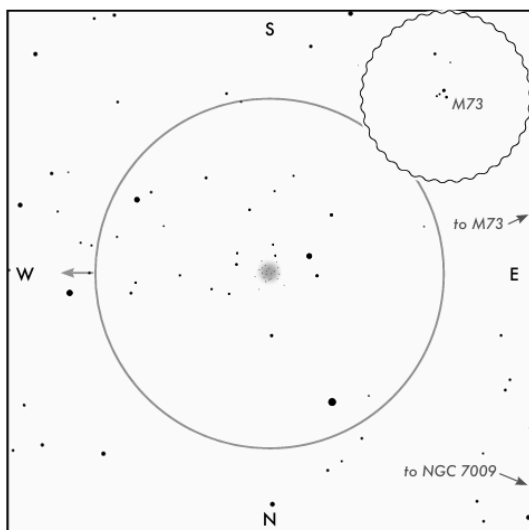
At low power, NGC 7009 is almost starlike, but with a distinct bluish-green hue, due to radiation emitted from doubly ionised oxygen created by the intense UV radiation emitted by the central white dwarf. The inset in the righthand image shows a high power (5') field of view of the Saturn Nebula, where short, faint equatorial ansae ('handles') can be seen to extend from the major axis, creating an impression of the planetary rings observed around Saturn. With no reference stars in the neighbourhood, the distance to NGC 7009 is not well known. Best estimates are of a few thousand years.



M72 (Globular Cluster) Hard ▾

RA 20h 54m 56s | Dec -12° 26' 07" | Mag 9.19

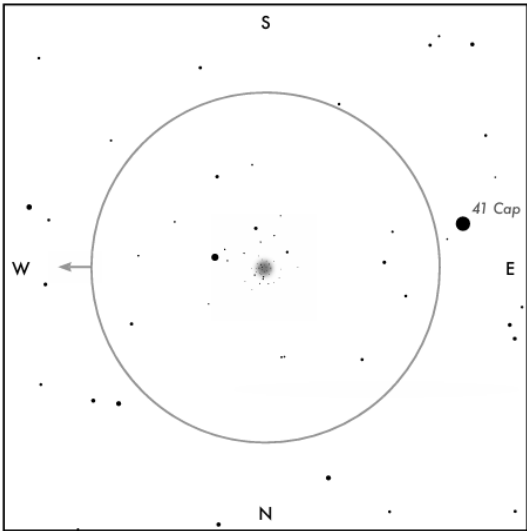
M72, one of the younger globular clusters, aged at 9.5 billion years, is loosely concentrated, with no sharp core, appearing as a pale, nebulous patch of light. Its fifteenth-magnitude stars are difficult to resolve, even with the large apertures of the Northumberland and Dobsonian. Distant 54,500 light years away, in a sparse region of the sky, M72 is a suitably difficult object to try star hopping to. Along the way, the collection of four, unrelated stars that constitute M73, may be located roughly 1.5° to the east of M72. M73, similarly to Brocchi's Cluster, was, at first, thought to be an open cluster.



In the constellation Aquarius, you might also like to observe:

- KV Aquarii (Variable, Double Star) Medium ▾
- M73 (Asterism) Hard ▾
- NGC 7293 - Helix Nebula (Planetary Nebula) Insane or Astrophotography ▾

Capricornus, Pegasus, Cepheus

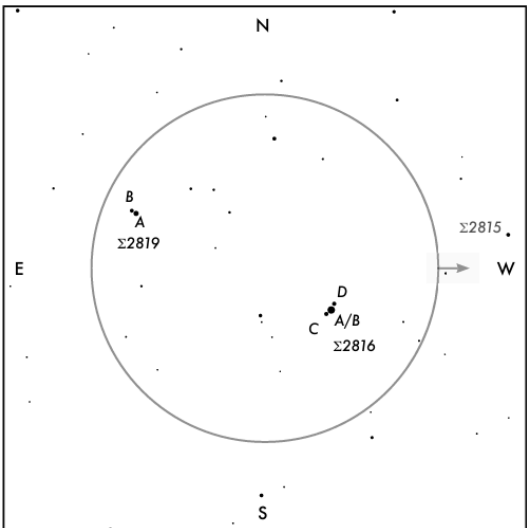
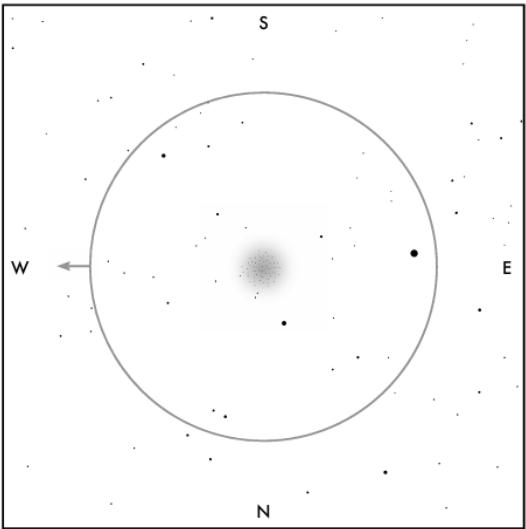


M30 (Globular Cluster) **Medium** ▼
RA 21h 41m 53s | Dec -23° 03' 26" | Mag 6.90

M30, positioned just below the ecliptic, is longest observable during the first half of August when it is positioned opposite the Sun. Rising to a maximum of ~ 15° above the horizon in Cambridge, it may still be a challenge to observe. In a low power field, M30 can be located as a dim disk to the west of the fifth-magnitude star, 41 Cap. The cluster extends to a diameter of 12', where individual stars may be resolved at high magnifications, and collapses to a core 1' wide with an extremely dense stellar population. With M30 distant 27,000 light years, its core measures less than one light year across.

M15 (Globular Cluster) **Easy** ▼
RA 21h 31m 16s | Dec 12° 17' 06" | Mag 6.30

As with M30, M15 has undergone a contraction, referred to as core collapse, resulting in a highly dense and bright nucleus that then thins rapidly from the centre. Higher magnifications will resolve individual stars, with dark lanes and patches prominent on the northeast edge of the cluster's inner shell. Distant 36,000 light years from Earth, M15 is also noted for being a source of X-rays (thought to be from supernova remnants, rather than a central black hole), and it contains the first planetary nebula, Pease 1, to be discovered within a globular cluster.



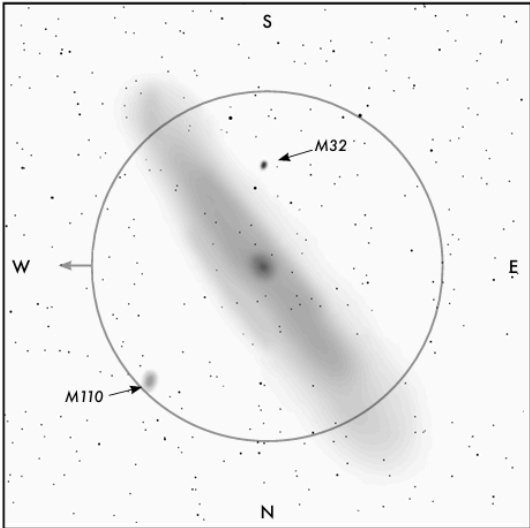
Struve 2816 and Struve 2819 (Multiple Stars) **Medium** ▼
RA 21h 39m 47s | Dec 57° 36' 35" (Struve 2816)

Star	Magnitude	Colour	Location
2816 A	5.73	Yellow	Primary
C	7.48	White	12" ESE from A
D	7.53	White	20" NW from A
2819 A	7.44	Yellow-white	Primary
B	8.64	Yellow-white	13" NE from A

In the constellations Capricornus, Pegasus and Cepheus, you might also like to observe:

- Mu Cephei - Garnet Star **Easy** ▼
- Delta Cephei (*The Cepheid Variable Star*) **Easy** ▼
- NGC 7317, 7318 A and B, 7319, 7320 - Stephan's Quintet (Galaxies) **Astrophotography** ▼

Andromeda



M31 - The Andromeda Galaxy Easy

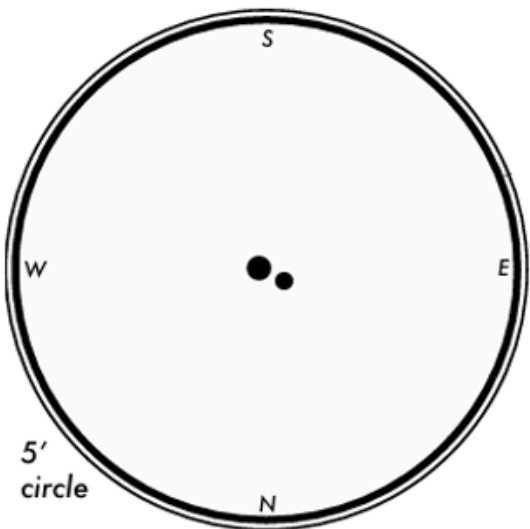
RA 00h 44m 12s | Dec 41° 24' 59" | Mag 3.40

Until 1925, when Edwin Hubble identified Cepheid variable stars within Andromeda, M31 was long believed to be a spiral nebula (as discussed in the 1920 ‘Great Debate’ between Shapley and Curtis). With 300 billion stars and a diameter of 150,000 light years, the Andromeda Galaxy is the largest of the Local Group. It is a barred spiral galaxy, like the Milky Way, observed nearly edge-on at a distance of 2.5 million light years. Observe the bright core of M31, allowing time for the surrounding galaxy to appear as a dim streak, extending far beyond the field of view of the lowest-power eyepiece.

M32 and M110 - Companion Galaxies of The Andromeda Galaxy Medium

RA 00h 44m 10s | Dec 41° 00' 48" | Mag 8.10 (M32) | | RA 00h 41m 50s | Dec 41° 49' 59" | Mag 8.10 (M110)

In 2006, using the Hubble Space Telescope, Grebel and Koch discovered that Andromeda’s faint companion galaxies lie close to a single plane through the core of M31. Independent interactions between galaxies should lead to randomly distributed satellites. One theory to explain this relative order includes a large-scale dark matter stream stretching from M33 to the M81 group! Alternatively, there is a possibility that a close encounter of a subhalo around the elliptical galaxy M110 with M31 could have contributed to the many corotating satellites. Only 12,000 light years across, M110 is of comparable brightness to the extended disk of M31, and is a peculiar galaxy in that it contains patches of dust and young blue stars near its centre. On the other side of M31 is the compact elliptical galaxy, M32, with an inner core of 330 light years. A similar galaxy has been observed in isolation without any massive companion to strip away its outer layers. Exactly how M32 formed is not well understood.



Gamma Andromedae - Almach (Double Star) Easy

RA 02h 05m 33s | Dec 42° 27' 32" | Mag 2.31, 5.02

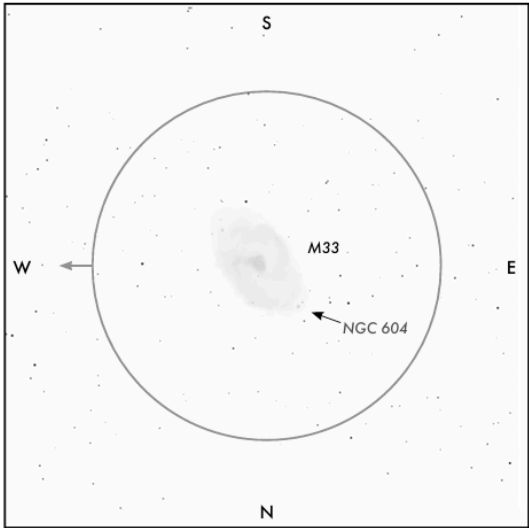
Star	Magnitude	Colour	Location
A	2.31	Yellow-orange	Primary
B	5.02	Blue-white	9.5" ENE from A

Distant 390 light years, Almach is a quadruple system consisting of a dying, class K3 giant (A) and a close pair (BC) of B8 and A0 main sequence stars orbiting at ~ 600 AU. The spectroscopic binary system of B has a 2.7 day orbital period.

In the constellation Andromeda, you might also like to observe:

- NGC 752 (Open Cluster) Easy
- NGC 7762 - Blue Snowball (Planetary) Nebula Medium
- Upsilon Andromedae (A Sun-like star with a planetary system that may be habitable.)

Triangulum



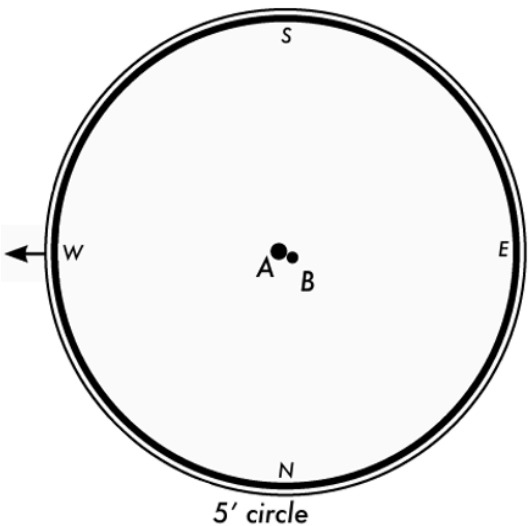
M33 - The Triangulum Galaxy Medium

RA 01h 35m 23s | Dec 30° 47' 46" | Mag 5.69

M33 is the third largest galaxy in the Local Group, after the nearby Andromeda Galaxy (M31) and our Milky Way. Though it is host to 40 billion stars, at a distance of ~ 2.9 million light years and with an apparent size of 71' × 42', the Triangulum Galaxy has an especially low surface brightness. This makes observations of it particularly difficult, and even more so dependent on the brightness of the surrounding sky – it is a galaxy that is not worth trying to view when the Moon is up! Despite being a spiral galaxy, M33 lacks the bright central core that can make such objects, like M31, easy to locate.

With dark-adapted vision and lowly light-polluted skies, M33 can be very impressive. Two loosely-wound spiral arms running west to north, and east to south form a discernible S shape in the nebulous patch, with several knots of light in the arms also being possible to locate. Most easily are the two that are ~ 10" south and southwest of the galaxy's centre, along with the supergiant H II region, NGC 604, positioned ~ 10" to the northeast. A cloud of hot, partially ionised hydrogen gas, NGC 604 is similar to the Orion Nebula (M42), though it is 6000 times more luminous. A nebula filter might help with viewing it.

Distant ~ 750,000 light years from the Andromeda Galaxy, it is thought that M33 may be a distant satellite. Water masers in the Triangulum Galaxy enabled astronomers to determine its angular velocity and proper motion (the angular speed of the object as it moves across the sky), and conclude that it is moving towards M31. Confirmed in 2011, a stream of hydrogen gas has been found connecting the two galaxies, which supports the idea that they may have tidally interacted in the past. A pure disk galaxy, M33 does not appear to have a supermassive black hole at its centre unlike M31 and the Milky Way.



6 Trianguli (Multiple Star) Medium

RA 02h 13m 56s | Dec 30° 25' 46" | Mag 5.26, 6.67

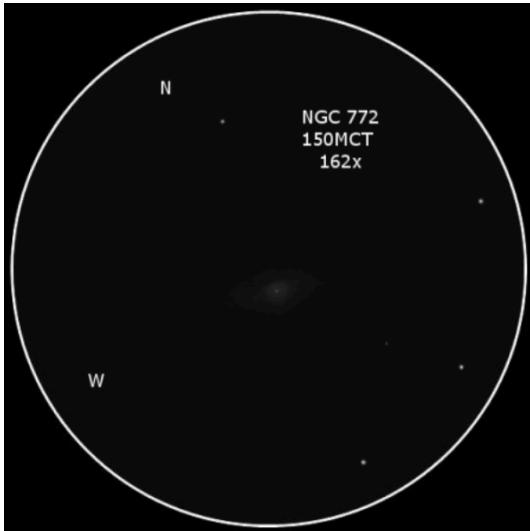
Star	Magnitude	Colour	Location
A	5.26	Yellow	Primary
B	6.67	Yellow-white	4.0" ENE from A

The visual pair, 290 light years distant, includes a G5III giant and its F6V companion. Both components are spectroscopic binaries, with very close orbits. The giant star produces a strong magnetic dynamo and has spots, like our Sun's.

In the constellation Triangulum, you might also like to observe:

- NGC 604 (Emission Nebula, in M33) Astrophotography
- (If the Triangulum Galaxy, M33, is not visible, it is best to move observations to another constellation.)

Aries, Pisces, Cetus



NGC 772 - Fiddlehead Galaxy Hard ▾

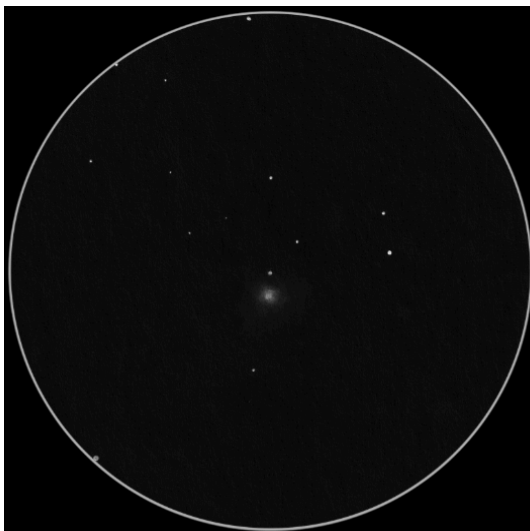
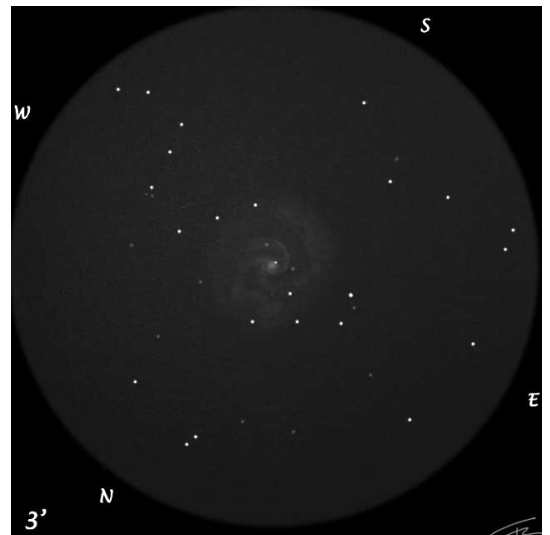
RA 02h 00m 48s | Dec 19° 08' 18" | Mag 10.30

NGC 772 is an unbarred spiral galaxy, distant 106 million light years. It has a diameter of over 200,000 light years, making it more than twice the size of the Milky Way. Likely inconspicuous even in a 12" Dobsonian, its asymmetric spiral arms are the result of tidal forces from its dwarf elliptical, satellite galaxy, NGC 770. The Michaelmas Star Hopping Guide directs the observer to the Fiddlehead Galaxy using the Dobsonian. It is best to begin observations at a moderate magnification, only increasing if the galaxy is bright enough. (Photo credit: Cloudy Nights 03/01/2016 Asbytec, 6" scope.)

M74 - Phantom Galaxy Hard ▾

RA 01h 38m 08s | Dec 15° 55' 14" | Mag 9.39

M74 (distant 32 million light years) is a face-on spiral galaxy 1.5° ENE of Eta Piscium, with two arms that widen as they wind anticlockwise from the centre. Whilst its large angular size and face-on orientation make it ideal for professional studies of spiral arm structure, they also result in M74 having the second-lowest surface brightness of any Messier object. It is best observed on a dark, moonless night at low magnification. When highly magnified, the diffuse emission is spread more widely and appears too faint to be observed. (Photo credit: Cloudy Nights 19/11/2020 FRANC LILL.)



M77 - Squid Galaxy Hard ▾

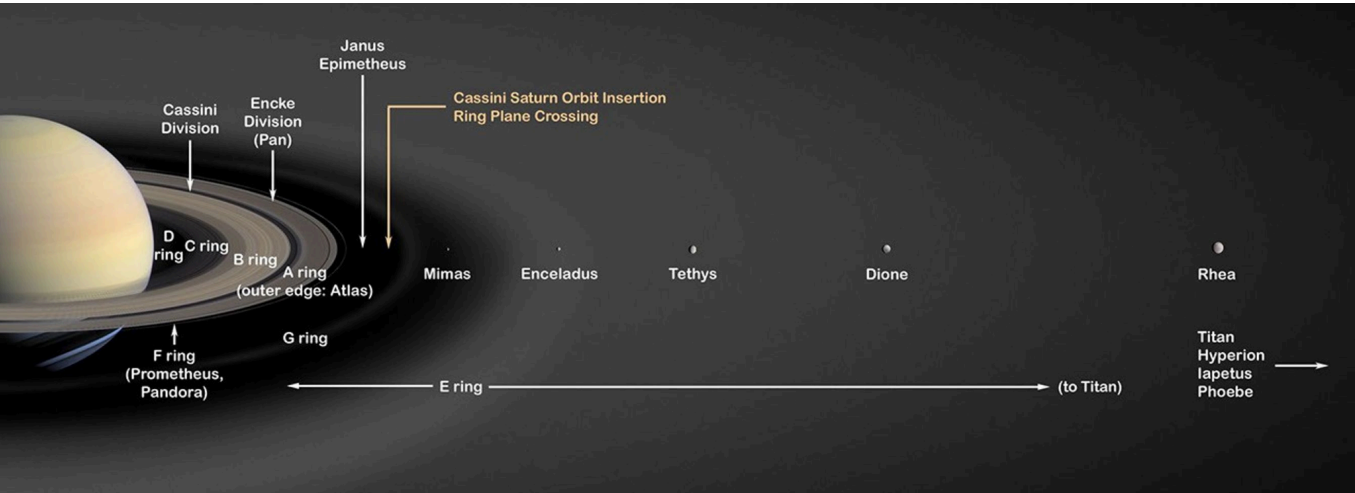
RA 02h 44m 03s | Dec 00° 06' 07" | Mag 8.89

One of the largest galaxies in Messier's catalogue, M77 is distant ~ 60 million light years away and can be easily found ~ 1° SE of Delta Ceti. The bright core of the barred spiral galaxy measures 120,000 light years across, and was found to be a strong radio source in 1952. A Seyfert galaxy, this radio source, Cetus A, has been found to be a supermassive black hole, orbited by a large disk of water molecules with an intense star forming region close to the nucleus. High power will reveal the starlike core and fragments of spiral arms. (Photo credit: <https://zvjezdopisi.com/>, 8" scope, Croatia.)

In the constellations Aries, Pisces and Cetus, you might also like to observe:

- Gamma Arietis - Mesarthim ((Visual) Double Star) Easy ▾
- 1 Arietis ((Unknown) Double Star) Medium ▾
- NGC 247 (Galaxy - a member of the nearby Sculptor Group; culminates at ~ 15°.) Hard ▾

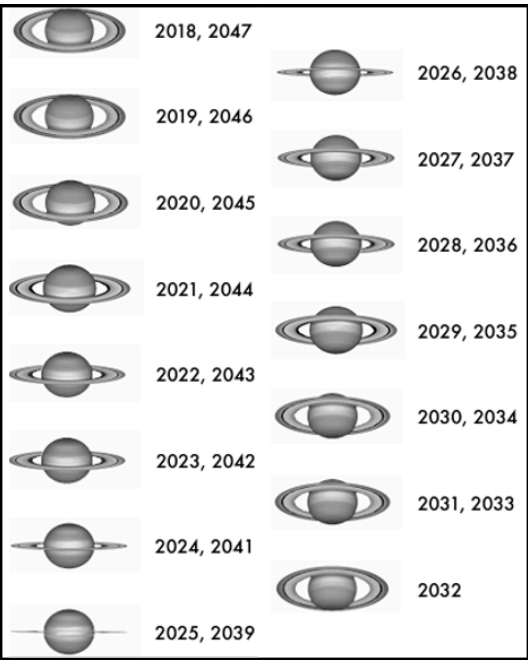
Saturn, in the constellation Cetus



Located in a fairly sparse region, on the border of the constellations of Pisces and Cetus, Saturn can be easily found as one of the brightest objects in the sky. When it reaches opposition on 04/10/2026, it will be comparable to the stars Vega and Capella, though as a planet with a visible disk, it should not twinkle. In the first half of 2025, Earth passed through the plane containing Saturn’s rings, meaning they are not visible to their full extent a year on, with only a fraction of their southern face being observable.

Plane crossings and the varying aspect of Saturn’s rings are the result of the planet’s 26.7° axial tilt relative to the normal to its orbital plane. When the line-of-sight vector from Earth to Saturn lies in the plane of the rings, they are seen edge-on, almost disappearing from view. This occurs every 13 – 15 years, roughly half a Saturnian year (29.4 years). Galileo’s accounts of these plane crossings from 1610 - 1613 are truly remarkable, bearing in mind he did not identify Saturn’s rings as such.

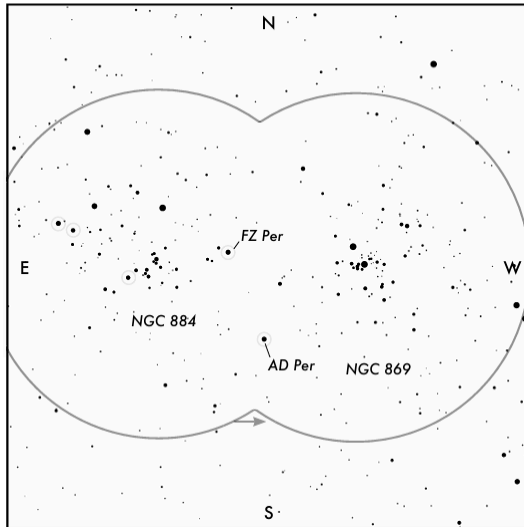
Though only a small fraction of Saturn’s rings will be visible in 2026, some of the dark divisions between them should still be discernible. The Cassini Division, a sharp, black line stretching half an arcsecond, separates the system into its two main components: the A and B rings. Look out, also, for the shadow the rings cast onto the planet. (Photo credit: NASA, *10 Things: Unsolved Mysteries of Saturn’s Moons.*)



The design of Saturn’s rings is largely the work of its many natural satellites, some of which can be readily observed. The second-largest moon in the Solar System, Titan, at eighth-magnitude, can usually be seen as a bright white dot near Saturn itself. With the rings near edge-on, 2026 is a favourable time to view the smaller and fainter (tenth-magnitude) moons that orbit closer to Saturn: Rhea, Tethys and Dione. Though unobservable, it is worth mentioning the moons Mimas, responsible for carving out the Cassini Division, and Prometheus and Pandora, which together act as shepherd satellites, maintaining the F ring.

A gas giant, Saturn’s atmosphere has the same banded structure as Jupiter, though further from the Sun, the cold and warm bands of ammonia ice are similar in temperature. This gives the surface a uniform appearance, with temperature being the distinguishing factor between it and Jupiter.

Perseus



NGC 869, NGC 884 - The Double (Open) Cluster

Easy ▾

RA 02h 24m 18s | Dec 57° 14' 22" | Mag 6.09

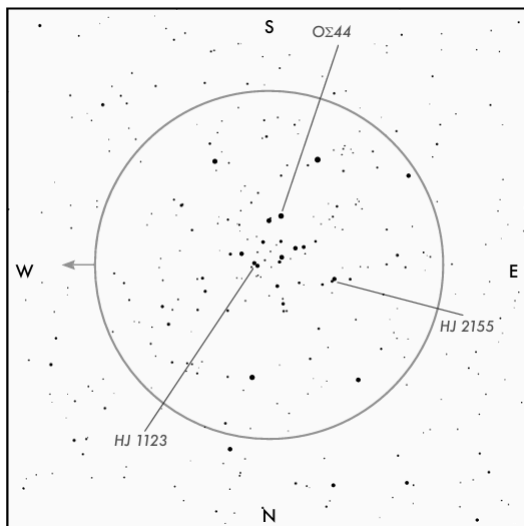
Observing both open clusters in a single field of view at low power is a spectacular sight. There are at least 5000 stars in the core of each cluster, with some (those circled in the lefthand figure) that are distinctly reddish. These are M2-class, red supergiants which, in a few million years, will undergo a Type II supernova. Both clusters are distant ~ 7500 light years away, though NGC 869 may be one or two hundred light years closer to us. At 14 million years old, a planetary system is unlikely to have formed in either cluster, but the view from one of the other would be unmatched.

M76 - Little Dumbbell (Planetary) Nebula

Hard ▾

RA 01h 44m 03s | Dec 51° 42' 16" | Mag 10.10

One of only four planetary nebulae to be recorded in Messier's catalogue, M76 is arguably the most difficult to observe given its low surface brightness and visual magnitude. Its name derives from the more impressive Dumbbell Nebula, M27, and is currently classed as a bipolar planetary nebula. This refers to the structure of M76, composing a ring seen edge-on that forms the central bar structure, with lower surface brightness lobes that extend from either opening of the ring. The progenitor star, upon exhausting its supply of hydrogen by nuclear fusion in its core, would have ejected the ring whilst evolving along the asymptotic giant branch (AGB). Soon afterward, the stellar core would have collapsed, expelling the remaining outer layers which, being partially blocked by the ring structure during expansion, would balloon to form the two lobes. M76 is best estimated to be 2500 light years away, though its gas expansion rate (and therefore its age) is not well known.



M34 (Open Cluster)

Medium ▾

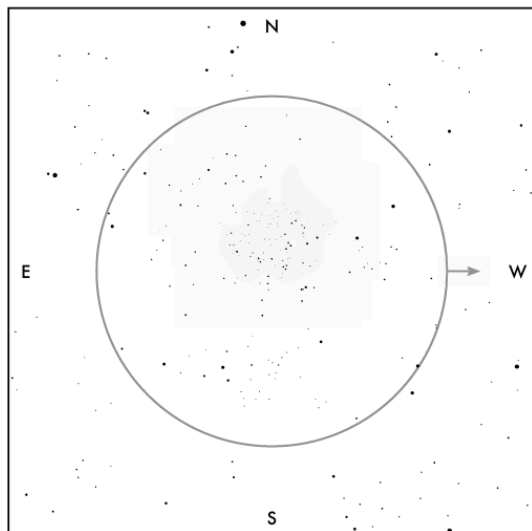
RA 02h 43m 49s | Dec 42° 51' 54" | Mag 5.19

Located 1500 light years away, M34 can be located just north of the line from Algol (Beta Persei, an eclipsing binary that varies in magnitude over a period of ~ three days) to Almach (Gamma Andromedae). From stellar spectra, M34 is estimated to be ~ 200 – 250 million years old, and is best observed at high power. This also allows for the challenge of splitting double stars present in the cluster, a few of which are labelled in the lefthand figure. Note that the Milky Way does pass through the constellation Perseus, so there will be a high density of background stars unrelated to the cluster.

In the constellation Perseus, you might also like to observe:

- Beta Persei - Algol (Variable, (Eclipsing) Double Star) Easy ▾
- Collinder 39 - Alpha Persei Moving Group (OB3 Stellar Association) Easy ▾
- AD Persei, FZ Persei (Variable, M2-class red supergiants) Medium ▾

Cassiopeia



NGC 7789 - Caroline's Rose (Open Cluster) Medium

RA 23h 58m 45s | Dec 56° 50' 59" | Mag 6.69

Distant more than 7000 light years, NGC 7789 is a large, rich open cluster with a prominent population of red giants. These G and K-class stars are the brightest members of the cluster and, being well-evolved, suggest it is 1.4 – 1.7 billion years old. With a total mass of $\sim 7000 M_{\odot}$, NGC 7789 is roughly an order of magnitude more massive than many of the open clusters recorded by Messier. At high powers, many of the stars will be resolved against a background haze, with loops of stars and dark lanes appearing as petals. Along with its discoverer, Caroline Herschel, this gives the cluster its name.

M52 (Open Cluster) Medium

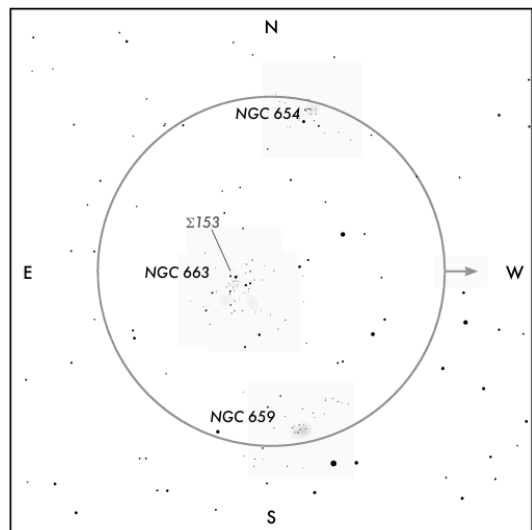
RA 23h 26m 00s | Dec 61° 43' 51" | Mag 6.90

Distant 5000 light years, M52 can be clearly resolved into 30 stars at medium power, and is dominated by an eighth-magnitude supergiant towards the southwest edge of the field of view. Obscuring interstellar matter between Earth and the cluster greatly affects its brightness and distance estimates.

M103 (Open Cluster) Medium

RA 01h 35m 10s | Dec 60° 47' 17" | Mag 7.40

A nebulous, wedge-shaped cluster, M103 is distant ~ 9000 light years and measures 15 light years across. At the centre, there is a distinctly orange star - the second-brightest member, a magnitude 10.8 red giant. A double star, Struve 131, appears in the foreground at the northwest point of the wedge.



NGC 654, 659, 663 (Open Clusters) Medium

RA 01h 48m 00s | Dec 61° 22' 05" | Mag 7.09 (NGC 663)

Within a low power field of view centred on NGC 663, two other open clusters (NGC 654 and 659) can be observed. Along with M103 and supergiant stars scattered between, these clusters are believed to form part of the stellar association Cassiopeia OB8. The stars share the same common origin, forming within a small volume of a giant molecular cloud, though are no longer gravitationally bound. The clusters, though separate objects, are all a few thousand light years distant and ~ 20 million years old. Observe individually at higher power, particularly NGC 663.

In the constellation Cassiopeia, you might also like to observe:

- NGC 457 - Owl Cluster (Open Cluster) Medium
- Iota Cassiopeiae (Multiple Star - three visual components) Medium
- NGC 281 - Pacman Nebula (Emission Nebula) Astrophotography

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